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9.3 IT Project Management

Mohammed Abdulwasa Mohammed Qasem

Contents

Abstract

In this report we will be covering comprehensive key areas of the project that has been delivered for the annual tournament system, this includes, the project's purpose, scope, and chosen methodology. Addition to that this report covers the project closure and reflection on how we managed the project, focusing on skills and behaviours that has been applied to, alongside there will be recommendations for future improvements.

Purpose

The purpose of the project was to develop an annual tournament system for Coventry Council, which allows the people of Coventry to participate as team or individuals to compete in five different events (Football, basketball, Tennis, Literature and Geography).

Scope

The project scope was to allow participants all ages to be part of the tournament, and the software that being used can allow the competitors to register as individuals or part of a team, able to lookup the competitions/events while competing on them and to see the Scoreboard. Additionally, the usability of the system must be easy and straightforward making it understandable and accessible for all users.

Chosen Methodology

The chosen methodology was Agile due to multiple reasons, compared to other methodologies like waterfall and RAD. Agile was the best fit and the closest to the project which allowed flexibility, iterative deployment and collaboration which made it ideal to use in this project, where the scope and requirements was needed to be refined based on client's feedback.

Project Management Skills Applied

In this project we used multiple skills and tools that helped us to track, update and manage the software product from start to finish. Such as, in the initiation phase we used **reverse thinking** to highlight the potential problems and provided insight into the overall system. For example,

1) *Instead of asking* "How can we design a tournament system that is efficient, user-friendly and engaging?"

   We can ask "How can we design a tournament system that is frustrating, inefficient and disengaging?"

2) *Instead of asking* "How can we design a fast, responsive interface with intuitive navigation?"
   We can ask "How can we ensure the system has a slow and unresponsive interface?"

Another key skill/tool we used was **feasibility study**, which we analysed the technical, economic, legal, schedule, security, sustainability, and operational aspects of the project. After conducting the feasibility study, we managed to identify the potential risks, constrains and benefits early in the planning process which enabled us to make informed decisions and resources efficiently such as using GUI user interfaces solution.

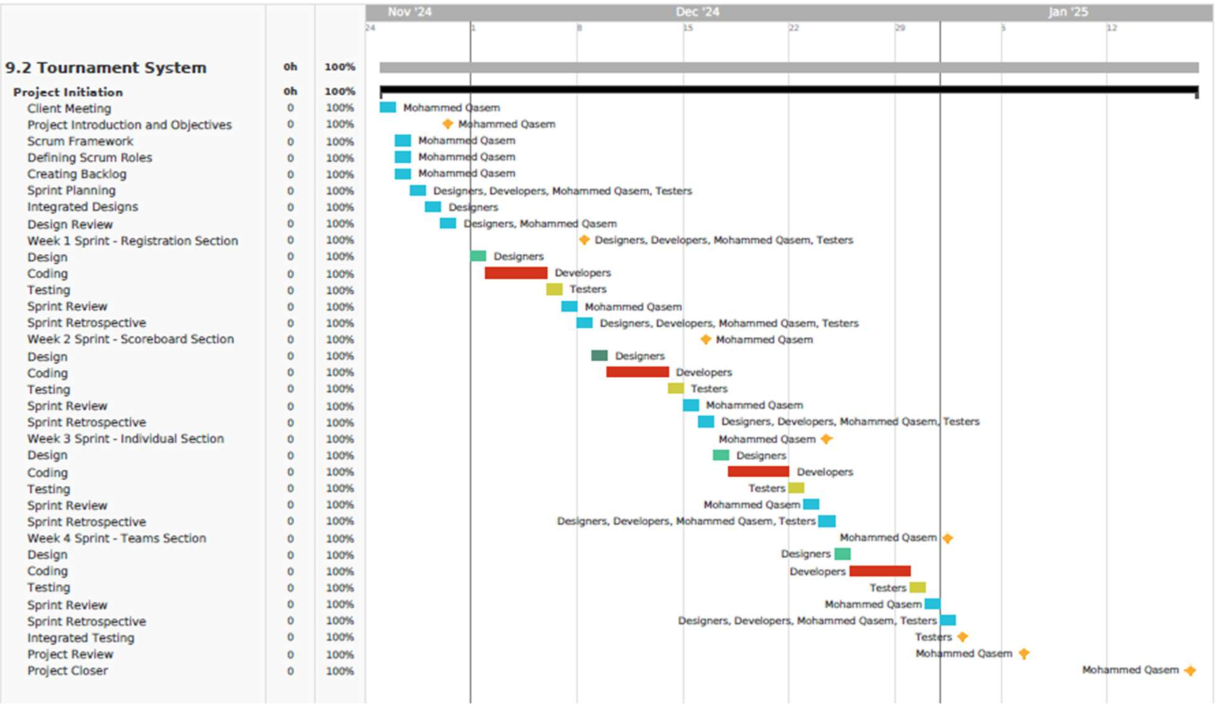
In the planning phase we used **Gantt Chart** to help us track progress, prioritise some tasks over other, allocate the available resources and make effective communication plan. This impacted us to put more efforts into Extreme, High, and medium tasks while putting less effort into low important tasks, which resulted us to save time and put high quality into valuable tasks.

Task's Title	Priority	Planned Start Date	Planned Finish Date
Initial meeting with the client	Medium	25/11/2024	25/11/2024
Analysing the requirements	Medium	26/11/2024	26/11/2024
Documenting the project	High	26/11/2024	26/11/2024
Analysing two alternative solutions	Medium	27/11/2024	27/11/2024
Feasibility study	High	28/11/2024	28/11/2024
Risk assessment	High	29/11/2024	29/11/2024
Chosen solution	Medium	29/11/2024	29/11/2024
Project phasing SRS	High	30/11/2024	30/11/2024
Project planning and monitoring	Medium	01/12/2024	01/12/2024
Risk and issue management	High	04/12/2024	04/12/2024
Analysing methodologies	Medium	05/12/2024	05/12/2024
Choosing a methodology	High	05/12/2024	05/12/2024
Dairy logs	Medium	05/12/2024	05/12/2024
Project introduction and objectives	Medium	07/12/2024	07/12/2024
Scrum framework	Medium	07/12/2024	07/12/2024
Defining scrum roles	Medium	07/12/2024	07/12/2024
Product backlogs	High	07/12/2024	07/12/2024
Sprint planning	High	08/12/2024	08/12/2024
Week 1 sprint	High	09/12/2024	15/12/2024
Week 2 sprint	High	16/12/2024	22/12/2024
Week 3 sprint	High	23/12/2024	29/12/2024
Week 4 sprint	High	30/12/2024	05/01/2025
Integrated test	High	06/01/2025	08/01/2025
Integrated design	Low	09/01/2025	10/01/2025
Others review	Low	11/01/2025	11/01/2025
Project reflection	High	11/01/2025	11/01/2025
Future improvements	Medium	13/01/2025	13/01/2025
Conclusion	Medium	13/01/2025	13/01/2025
Project delivery	Extreme	14/01/2025	17/01/2025

In the execution phase we used tools and methods like, **backlogs** which is the user stories and requirements needed to be developed in each sprint by identify the user stories were able to know what features are needed to be developed in each sprint cycle, **Moscow** which helped us to determine the level of importance for backlogs, **sprint planning** which includes design/develop/test/sprint review/sprint retrospective due to Agile being chosen methodology in this project.

Additionally, in this phase we also conducted **integrated testing** which we tested all the features of software application system we managed to spot earlier any bugs or issues after development phase, we used **Microsoft form** to collect feedback reviews from others for user interface design, and lastly, we created **dairy logs** to record all the changes that has been made based on the client's feedback. With all these tools we managed to effectively carry out efficient testing, positive feedback on the interface design and we had good tracking on the changes that have been asked by the client.

In monitoring and controlling phase, we used tool like **Team Gantt** for monitoring and tracking progress. Which recorded each sprint activity with the team member that was assigned to it. For example, designers were asked to design the user interface of each feature window and send it to other colleagues for improvements purpose, while developers start executing the user stories of the current sprint cycle and developing the coding bit of the sprint. By combining all this tools/method we managed to effectively execute all the tasks that were needed to be done in each sprint and do final testing.



Project Management Behaviours Applied

In this project we also used soft skills and professional behaviours such as, active listening during meetings, this skill was useful because it allowed the client and stakeholders to speak freely without us interrupting them, to let us make notes and know clearly what we need to do after the meeting. Because we applied active listening, we managed to avoid misunderstandings requirements and reducing rework.

We used time management effectively, by having good planning and tracking progress which allowed us to even finish the whole project 2 weeks before the deadline, which we took it as advantage to double check everything that is related to issues/bugs in the software product. I ensured to complete many tasks each week for example, the first week we decided to only observe and analyse the user requirements/stories to have an idea on how we will start the project, we then used the second week to conduct feasibility studies and risk assessments. Additionally, we chose GUI as suitable solution based on the feasibility result.

Furthermore, we used good collaboration skill that allowed us to share and collect effective feedback from others about the user interface design via Microsoft form, and working closely with the client and stakeholders to refine or confirm the developed features in each sprint cycle. For example, when the client asked to refine “edit and “remove” buttons in the user interface GUI, we managed to do it on time and the result was positive to the client.

Areas for Improvement

Report section

In the report section, the feasibility study had minor gap in recommending the solution based on feasibility study, it could have been more in depth and using example to compare the available resources to have better logical chain of reasoning to conclude the adopted solution, by doing this the readers of the report can have more in-depth reasonable thinking on why we chose GUI instead of other solution like CLI.

Also, during the execution phase, I have added a snippets of project code script without mentioning briefly about those snippets/modules which could have helped the readers to have a clear idea of the code and why we developed it this way.

Additionally, as future refine approach, we will use python libraries such as, NumPy or Panda which could have helped our developers to perform various tasks without writing code from scratch, to simplify development and implementing complex functionality efficiently. This method could had benefit in terms of saving more time which we could have invested it in something else like more quality assurance.

Software product

The final software developed product was able to run on desktop devices such as, laptops and PCs only. Which in the future improvement we will ensure to include other devices like, smartphones and tablets to be able to access the application and allow them to participate in the tournament. This way we could had more users than now because many users do not have access to PCs and laptops device to allow them to participate in the tournament.

Furthermore, maintaining a high standard of security is most important thing to protect the system and the organisation’s data. Which the current system does not have high security standards to protect the system. So, it is crucial to implement a robust measure such as, incorporating firewalls into the internal network infrastructure to guard against unauthorised access.

Installing and regularly updating antiviruses software on all devices, both old and new, will help safeguard against malware and other cyber threats. To complete these measures, routine security testing of devices should be carried out to detect and mitigate any malicious activity promptly. By prioritising these security enhancements, the organisation can minimise vulnerabilities and reduce the risk of cyberattacks.

Overall Reflection

This project was successful in terms of meeting all the client requirements and user stories. The system allows the people of Coventry to participate in 5 different events, which ranges from sport to academics. The system also, have validation rules to not allow certain things to happen or to be exceed such as, not allowing to register more than 4 teams and each team cannot have more than 5 players. All the features and functionality work efficiently and there is no bugs or issues with the final software product.

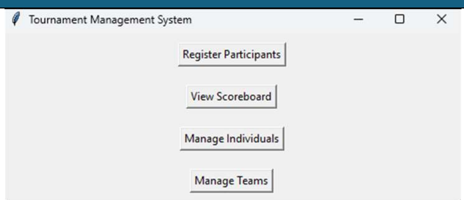
The project objectives were:

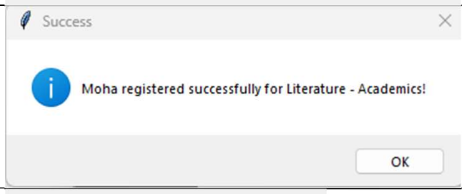
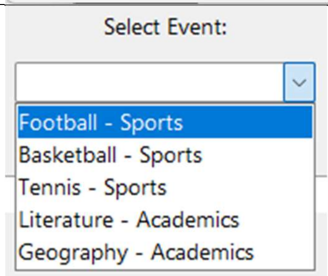
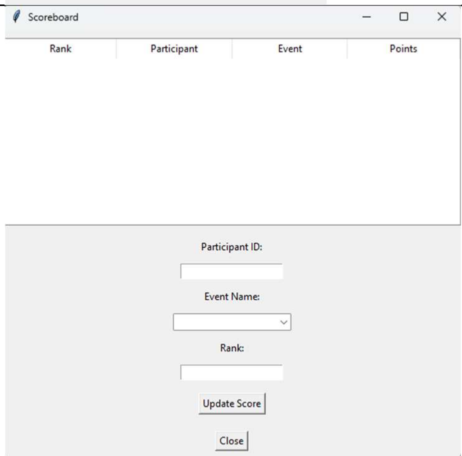
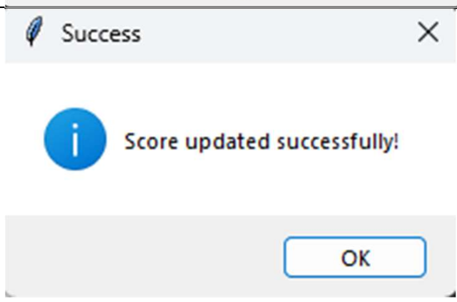
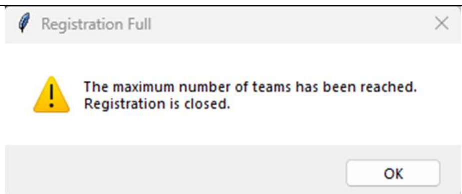
- Users can register as part of a team or individual **(M)**
- There are only 20 slots available for individuals **(S)**
- There are 5 different events in the tournament **(M)**
- Users can view the scoreboard **(S)**
- Users can input their ranking that they end up with on events **(M)**

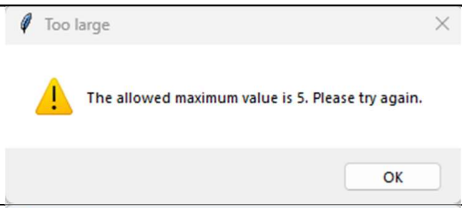
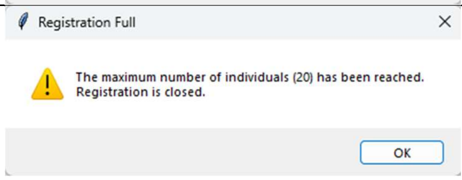
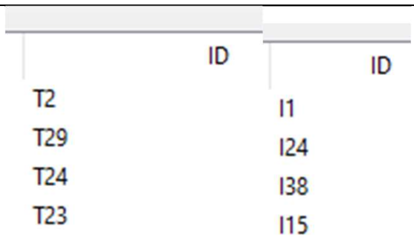
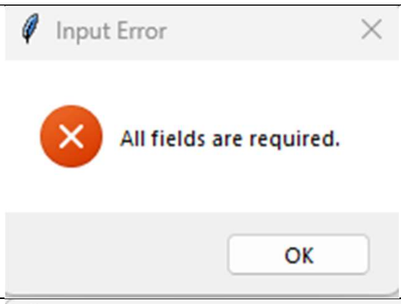
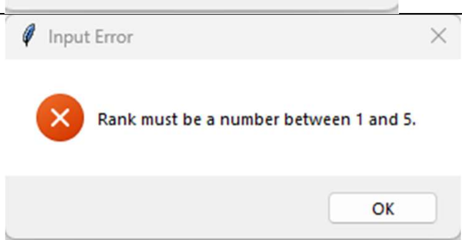
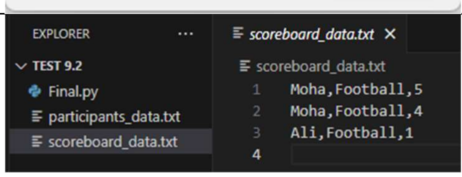
To make the system more effective and accurate, we will include some validation rules to ensure that the system runs efficiently and record data accurately. By include these rules:

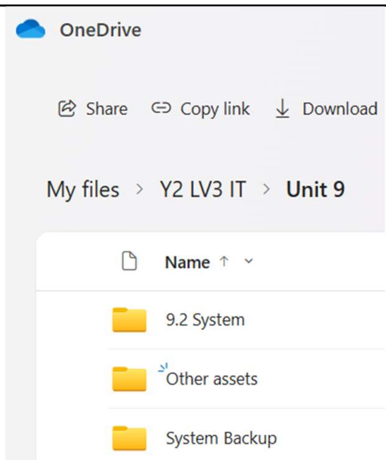
- Users cannot register more than 4 teams **(C)**
- Users cannot register more than 5 players in a team **(C)**
- Users cannot register more than 20 individuals in all events **(C)**
- Users cannot leave or input incorrect information **(C)**
- No duplicate IDs **(C)**
- Admins can manage teams, individuals, and scoreboard windows **(C)**
- Data must be securely stored in secure external file **(C)**
- There must be backup option in case of accidental deleting/failed system **(C)**

The test result outcome:

Test No	Test Description	Evidence	Pass/ Failed
1	Starting the program without any issues/bugs		Pass

2	Registering a team consisting of 5 players		Pass
3	Registering an individual		Pass
4	Open menu to let users select the event they want to register		Pass
5	All users can view the scoreboard		Pass
6	Users can input their ranking in the scoreboard window. (There is a dependency on their IDs for this test. To input their rank, users must provide their ID. Without the ID, they cannot complete this action)		Pass
7	Users cannot register more than 4 teams		Pass

8	Users cannot register more than 5 players in a team		Pass
9	The maximum slots available for individuals is 20. (The system will not allow any more registrations if the maximum number of slots has been reached)		Pass
10	The system should not give any duplicate IDs. (Every team and individuals must have unique IDs)		Pass
11	Users cannot leave any field empty when registering or inputting their rank		Pass
12	Users cannot input wrong rank when updating the scoreboard.		Pass
13	When users input their rank, the system will save the data to an external file to ensure that the ranks are securely stored outside the system		Pass
14	When users input their rank, the system will automatically assign points based on the pre-set scoring system		Pass

15	Admins have full access to the system, including the ability to backup data from cloud storage. This ensures that in the event of a system failure or data loss, a backup is available		Pass

Major lesson learned.

One major lesson learned was that providing clear documentation for code snippets and system configurations from the start would have improved collaboration and clarity. Some team members struggled to understand previously written functions due to lack of comments and documentation.

Critical analysis

We used weeks deadline strategy as measure of achieving project goals, meeting stakeholder expectations, delivering value, and achieving a positive return on investment (ROI). Such as, monitoring each tasks progress from (Extreme to low) to ensure we put the most effort on the extreme and high tasks to have effective outcomes rather than putting all the efforts into low tasks which could have resulted us in wasting time and negative client satisfaction rate.

Due to the project deadline was near, we prioritised developing the core tournament registration features first. As a result, we had limited time to refine the user interface, leading to a less polished design in some areas.

We also, used project management terminology such as, risk matrix and iterations to have best outcomes of the project. The use of risk matrix helped us to identify and categorise the different potential risks we could face including the severity impact of that risk. We used contingency plan to also, identify the potential risks and produce measurable mitigation strategies to tackle all the issues that can delay the project or make it fail.

Furthermore, the iterations were part of our chosen methodology (agile) which helped us to have iterative cycles to be designed/developed/tested/reviewed and based on the client's feedback we determined which features to confirm or refine. Alternatively, using RAD as replacement could also be suitable in this type of projects that needs user and stakeholders' involvements and based on their feedback you can refine it, unlike waterfall which in this type of projects it is not suitable due to rigid nature of not allowing changes in later stages.

Conclusion

Overall, this project was successful in terms of meeting all the client requirements, quality assurance, user stories, risk mitigation, low budget and meeting the delivery deadline. Along with creativity and suitable solution we managed to satisfy the client and stakeholders of the final project. Keeping in mind both us and the client needs to carefully implement the suggested improvements into future projects to ensure high quality product.